

*Effects:

- (a) Blurred or distorted vision
- (b) A point is seen as a line & a line appears to have a halo on either side
- (c) Astigmatism may be myopic, hyperopic or both

*ttt: Cylindrical lens, with its curvature // to the plane to be corrected (?)

Lecture (19): The Visual Process

(Dr. Wagdy Nashaat Habeeb)

1. Duplicity Theory of Vision

“2 types of photoreceptors (rods & cones) → 2 separate mechanisms of vision (scotopic & photopic)”

	Rods	Cones
(A) <u>Anatomical</u> :		
1. Number:	120 million in each retina	6 million in each retina
2. Site:	Peripheral (mainly)	Central (mainly) “Fovea centralis contains cones only”
3. Photopigment:	*More *1 type (rhodopsin)	*Less *3 types (blue, green & red)
4. Convergence:	200:1 *200 rods → 1 bipolar cell *Many bipolar cells → 1 ganglion cell	1:1 (no convergence) *1 cone → 1 bipolar cell *1 bipolar cells → 1 ganglion cell
(B) <u>Physiological</u> :		
1. Light sensitivity:	Very high (Up to 300 times as cones)	Very low
2. Visual acuity:	Very low	Very high
3. Color perception:	No	Very high
4. Vision:	Scotopic (dim / night) vision	Photopic (bright / day) vision

2. Photochemistry of Vision

➤ Phototransduction:

*Definition:

It is process by which light energy is converted into receptor potential in the photoreceptors

*Mechanism (Sequence of events):

- (A) Photochemical changes “Bleaching”
- (B) Genesis of electrical response in retina
- (C) Excitation cascade
- (D) Signal transmission in retina

(A) Photochemical changes “Bleaching”:

*Photopigments: 2 types;

1. ***In rods:*** → 1 type;

- “Rhodopsin”: Called visual purple as it has a purple color on absorbing green or blue light
- Formed of 11-cis retinal & an opsin protein (scotopsin)

2. ***In cones:*** → 3 types (each cone contains only 1 type);

- “Cyanopsin”: blue-sensitive

“Iodopsin”: green-sensitive

“Porphyropsin”: red-sensitive

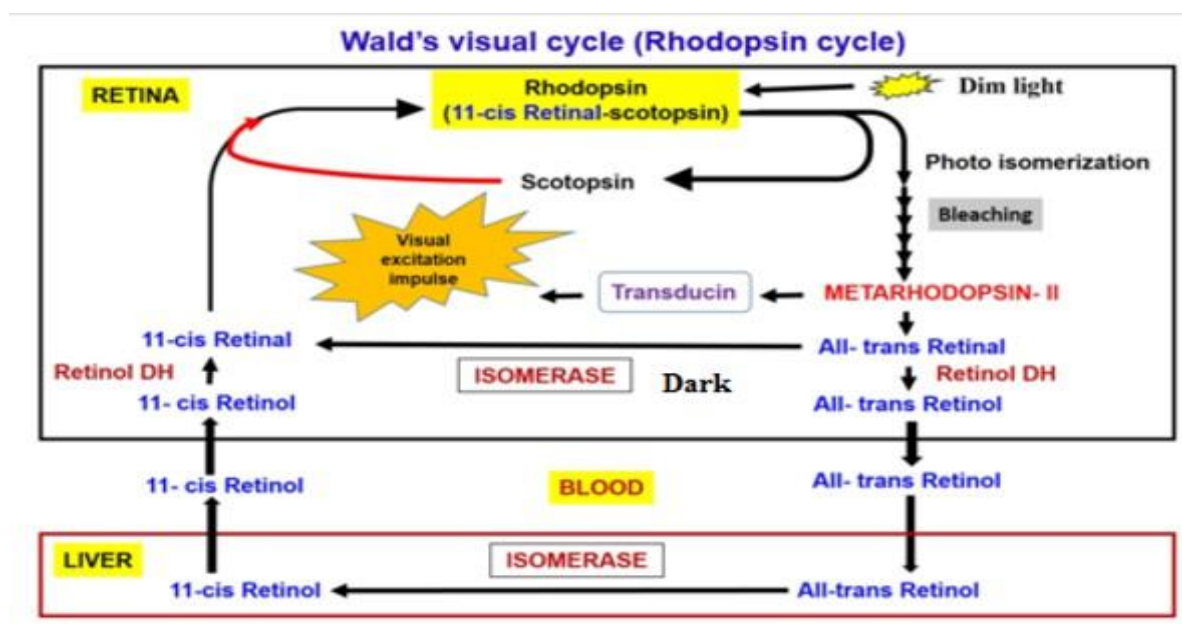
- Formed of 11-cis retinal & an opsin protein (photopsin; 3 types)

N.B. Retinal is the aldehyde of vit. A₁ (retinol)

***In Rhodopsin:** “Wald’s visual cycle”

On light exposure: Rhodopsin immediately decomposes

In the dark: Rhodopsin is regenerated



***In cone pigments:**

Similar changes occur, but regeneration is much more rapid than rhodopsin

Role of vitamins in retinal function:

***Vit. A:** is essential for synthesis of 11-cis retinal part of the photopigments

Vit. A deficiency: →

1. Nyctalopia (night blindness): due to failure of dark adaptation
2. Degeneration of cones

If prolonged → 3. Degeneration of the neural layers of retina

4. Xerophthalmia

***Vit. B₂**: is essential for normal retinal function as nicotinamide (vit. B₅) is a part of NAD which acts as a coenzyme for retinol dehydrogenase in rhodopsin cycle

(B) Genesis of electrical response in retina:

***Dark (during rest)** → Depolarization “RMP = – 40 mV (low)” → allows continuous release of synaptic transmitter (glutamate) → (–) bipolar cells

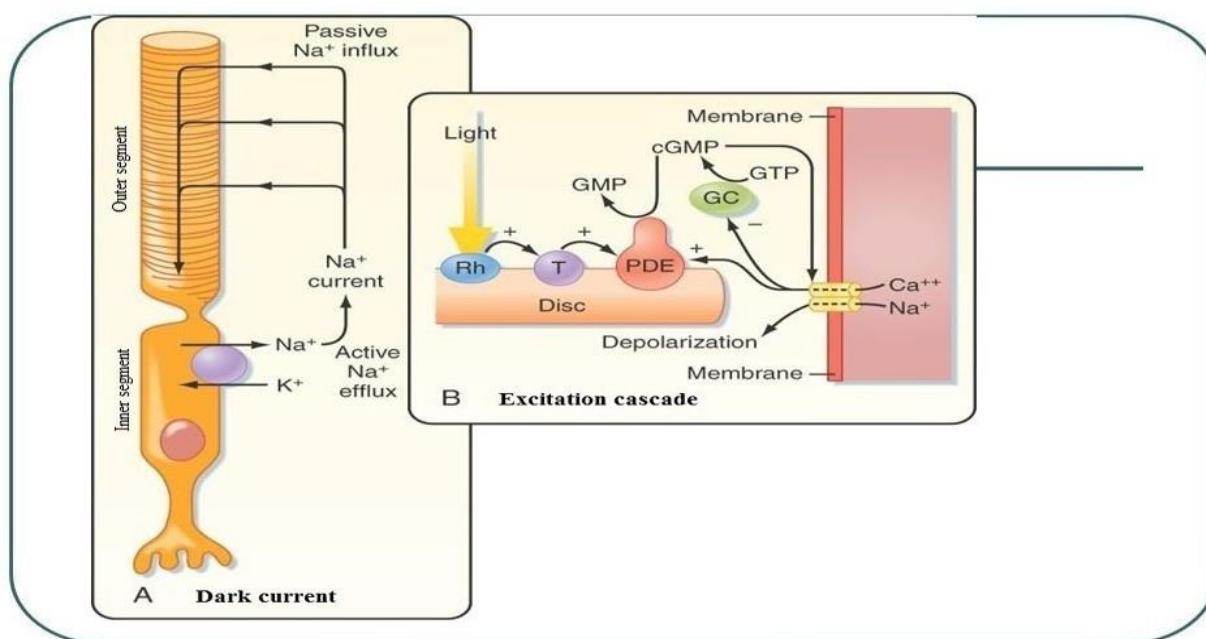
***Mechanism:** Dark current (Figure A)

(C) Excitation cascade of photoreceptors:

***Light exposure** → Hyperpolarization “RP may reach – 70:– 80 mV (directly proportionate to logarithm the intensity of light according to Weber-Fechner’s law)” → ↓↓ release of synaptic transmitter (glutamate) → no (–) of bipolar cells → depolarization → AP in ganglion cells

***Mechanism:** Excitation cascade (Figure B)

N.B. RP of rods lasts for a longer time than cones (?)



(D) Signal transmission in retina:

- Electrical response of retinal neurons is RP (local graded potential) & not AP → Conduction of visual signals by electronic conduction → allows graded conduction of signal strength

N.B. Exceptions are ganglion cells & some amacrine cells which conduct signals as AP

Lecture (20): Light-Dark Adaptation & Color Vision

(Dr. Wagdy Nashaat Habeeb)

1. Retinal Adaptation (Automatic Regulation of Retinal Sensitivity)

➤ Definition:

Ability of retina to adjust its sensitivity to different light intensities

➤ Types:

	Dark adaptation	Light adaptation
1. <u>Definition:</u>	↑↑ retinal sensitivity to light (by 100.000 to 500.000 times), <i>i.e.</i> , ↓↓ retinal threshold	↓↓ retinal sensitivity to light, <i>i.e.</i> , ↑↑ retinal threshold
2. <u>Occurs when:</u>	Person shifts from brightly-lighted place to dark place	Person shifts from dark place to brightly-lighted place
3. <u>Timing:</u>		
(a) Starts:	After 1 min	After 1 sec
(b) Reaches max.:	Within 30 min	Within 5 min